

GEYI YANG

123090721@link.cuhk.edu.cn | GaryYang | +86 13725353364

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A1b06, Minerva College, The Chinese University of Hong Kong, Shenzhen
2001 Longxiang Rd, Longgang Dist., Shenzhen, Guangdong, China

EDUCATION

- The Chinese University of Hong Kong, Shenzhen** Sep. 2023 – Present
B.Eng. in Computer Engineering Shenzhen, China
 - GPA: 3.73/4.00.
 - Selected coursework: Artificial Intelligence (A), Machine Learning (A), Optimization (A), Data Structures (A-), Operating Systems (A-), and Internet Architecture and Protocol (A-).
- University of California, Berkeley** Jan. 2026 – Jun. 2026
Exchange Student in Computer Science Berkeley, CA, USA

HONORS AND AWARDS

- Mathematical Contest in Modeling (MCM), Finalist (F Award; top 1%)** 2026
- Academic Performance Scholarship Awardee, CUHK-Shenzhen** 2023–2024
- Dean's List, CUHK-Shenzhen** Three consecutive years

RESEARCH EXPERIENCE

- HarnessGrad: Adaptive Harness Optimization for GUI Agents** Jul. 2026 – Present
Advisor: Prof. Zhongxiang Dai, School of Data Science, CUHKSZ 
 - Study inference-time harness optimization for GUI agents, targeting accuracy-cost-latency trade-offs and Pareto-efficient agent configurations without changing the underlying models.
 - Designed a recursive self-improving (RSI), rollout-failure-driven loop: collect actions, screenshots, execution feedback, and rewards; identify systematic failures; attribute them to editable harness components; and generate candidate updates.
 - Built a closed loop of trace analysis, failure clustering, harness attribution, paired retesting, cost profiling, and retain-or-rollback decisions; jointly evaluate success rate, execution time, and tool/API costs.
 - On OSWorld with fixed Qwen3-VL-32B, UI-TARS-1.5-7B, and Agent-S3 components, improved the optimization-set score from 0.267 to 0.454 (+70%); an independent rerun reached 0.464.
- Chinese Internet Meme Conversational Language Model** Jan. 2026
Dataset Construction, Fine-Tuning, and Preference Alignment  Model |  Dataset
 - Independently built and open-sourced the full data-to-model pipeline: collected 30,000+ post-comment pairs with Playwright and released an 8,680-example Chinese meme SFT dataset in ChatML format.
 - Fine-tuned Qwen2.5-7B on a single A100 (40GB) with LoRA and DPO; improved meme-use accuracy by 32%, with human and LLM preference rates of 82%/91%/68% for humor, accuracy, and naturalness.

INTERNSHIPS

- China Academy of Sciences Cloud Computing** Mar. 2025 – Jun. 2025
Research Intern – Agentic-RAG Question-Answering System for Smart Mining Dongguan, China
 - Built an Agentic-RAG system for mining-blasting technical documents using Python, LangGraph, LangChain, Qdrant, and FastAPI; implemented planning, ReAct, and human-in-the-loop components.
 - Developed Docling parsing and parent-child hybrid retrieval (BGE-M3 + BM25 + BGE reranking), achieving Recall@5 of 81% and answer accuracy of 78% on 300+ QA samples.
- Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS)** Sep. 2023 – Jun. 2024
Research Intern – Digital Human Synthesis from Image and Speech Shenzhen, China
 - Built a multimodal data pipeline that processed historical photos and audio recordings into visual and speech datasets for voice cloning and image-driven synthesis.
 - Integrated MockingBird voice cloning, SadTalker lip-sync and facial animation, and LLM-based dialogue to create an emotionally responsive “Digital Relative” interaction system.
 - Delivered an end-to-end prototype piloted with family users: completed 500+ interactions, with 83% of participants reporting a familiar emotional connection and 20% reporting strong emotional resonance.

TEACHING EXPERIENCE

- **The Chinese University of Hong Kong, Shenzhen**
Undergraduate Student Teaching Fellow (USTF) – MAT1001 Calculus I
 - Served as a USTF for the first-year calculus course MAT1001, supporting tutorials and student learning.

Sep. 2025 – Dec. 2025
Shenzhen, China

SELECTED PROJECTS

- **Telos: AI-Generated, Ready-to-Learn Courses**
Independent Full-Stack Developer
 - Developing a goal-driven course-generation system that transforms fragmented user materials and online resources into structured, multimodal, ready-to-learn courses.
 - Designed the output as a polished learning workspace with embedded text, images, and videos; planning course editing, AI tutoring, interactive exercises, flashcards, and multidimensional retrieval.
 - **TaskFlow AI: Visual Task Orchestration and Automation**
Independent Full-Stack Developer
 - Built a dual-level node-flow workspace for visualizing tasks, dependencies, statuses, deadlines, and serial/parallel/merge sub-workflows.
 - Integrated OpenClaw and Lark to extract tasks from email, create and advance workflows, request confirmation through message cards, and synchronize reports, tables, and calendars.
 - **Voice Turing Test Data Collection Platform**
Full-Stack Development Lead
 - Led full-stack development of a WeChat Mini Program supporting research published at ICLR 2026; implemented experiment flow, bilingual UI, real-time scoring, audio preloading, and caching.
 - Integrated Tencent Cloud services and designed the UI/UX in Figma; added retention features that acquired 1,000+ users, reached 300+ DAU, and collected 5,000+ valid records in the first month.

Aug. 2026 – Present

Apr. 2026

May 2025 – Aug. 2025

SKILLS AND LANGUAGES

Programming & Systems: Python, C++, JavaScript, TypeScript, Shell, FastAPI, React, SQLite, Git, Linux.
Machine Learning & Agents: PyTorch, Transformers, LoRA/PEFT, TRL (DPO), RAG, LangChain, LangGraph, GUI agents, Qdrant, MCP.
Languages: English (TOEFL 111; GRE 332); Mandarin Chinese (native).